

Pediatric Hospital Throughput Analytics Demo

Microsoft Fabric | Power BI | Dataflow Gen2 | Lakehouse | Healthcare Operations Analytics

Project Overview

This project demonstrates an end-to-end healthcare analytics workflow using 1,000 synthetic pediatric encounter records. The workflow shows how raw encounter data can be ingested, transformed, validated, modeled, and presented in an executive-facing Power BI report.

The data is synthetic only and does not contain patient-identifiable information, real hospital records, protected health information, or proprietary data from any healthcare organization.

Executive Metrics

Metric	Value
Total Encounters	1,000
Average LOS	4.27 days
Emergency Admission %	76.2%
Long Stay Rate	31.9%
30-Day Readmission Rate	5.7%

Workflow

Raw CSV source -> Fabric Lakehouse Bronze table -> Dataflow Gen2 transformations -> curated Gold table -> Power BI semantic model -> executive dashboard.

Transformations included age group creation, admission type standardization, discharge status mapping, encounter month creation, length-of-stay categorization, long-stay flagging, readmission labeling, and source-to-report validation checks.

Key Findings

- Emergency Department encounters represented the largest department volume.
- The overall average length of stay was 4.27 days.
- Emergency admissions accounted for 76.2% of encounters.
- Long stays of 4 or more days accounted for 31.9% of encounters.
- The 30-day readmission rate was 5.7%.
- NICU and PICU had the highest average length of stay, while the Emergency Department had the lowest average LOS.

Power BI Report Snapshot

1K

Total Encounters

4.27

Average LOS

76.2 %

Emergency Admission %

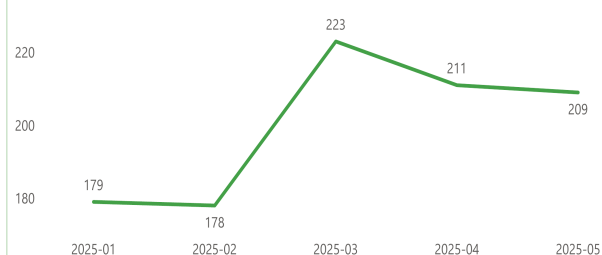
31.9 %

Long Stay Rate

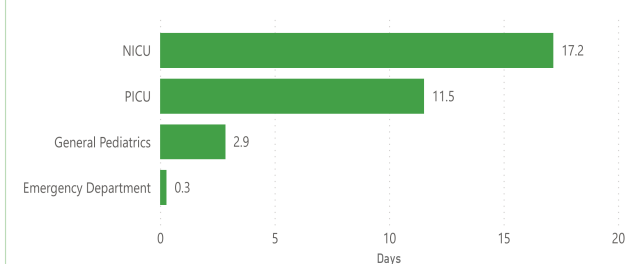
5.7 %

30-Day Readmission Rate

Monthly Encounter Volume

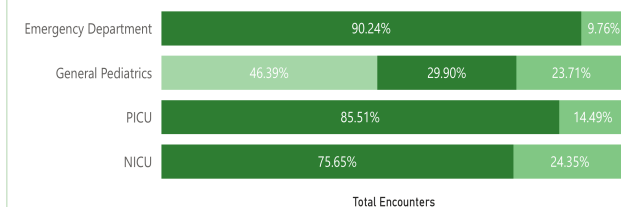


Average LOS by Department

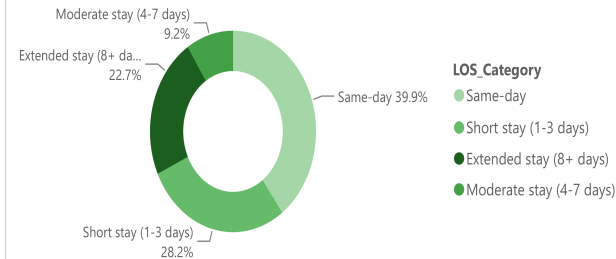


Admission Type Mix by Department

Admission_Type ● Elective ● Emergency ● Urgent



Encounter Distribution by LOS Category



Fabric Workflow Snapshot

Microsoft Fabric: Pediatric Encounter Demo

Synthetic pediatric encounter data. Built to demonstrate ingestion, transformation, semantic modeling, validation, and Power BI reporting.



1. Bronze Layer

- Workspace: CHOA ANALYTICS DEMO
- Lakehouse: Pediatric_Data_Lake
- Raw table: pediatric_encounters_bronze
- Loaded source encounter file into Lakehouse

- Purpose: preserve source structure**

Pediatric_Data_Lake
Tables
dbo
pediatric_encounters_bronze
ABC Patient_ID
ABC Encounter_ID
123 Age
ABC Age_Group
ABC Admit_Date
ABC Admit_Month
ABC Admission_Type
ABC Department
123 Length_of_Stay_Days
ABC Discharge_Status
123 Readmit_30_Day_Flag

2. Transformation Layer

- Applied Dataflow Gen2 transformations
- Created derived reporting fields
- Prepared data for validation and modeling

- Purpose: shape raw data into analytics-ready structure**

Query settings
Properties
Name: pediatric_encounters_bronze
Applied steps
Source
Navigation
Navigation 1
Navigation 2
ABC Changed column type
ABC Inserted conditional column
ABC Added custom
ABC Added custom 1
ABC Added custom 2
ABC Added custom 3
ABC Reordered columns
ABC Changed column type 1

3. Validation

- Verified total encounter counts
- Checked department level counts
- Confirmed data types and reviewed derived reporting fields

- Purpose: validate source-to-report accuracy**

```
1 SELECT
2   Department,
3   COUNT(*) AS encounter_count
4 FROM pediatric_encounters_bronze
5 GROUP BY Department
6 ORDER BY encounter_count DESC
```

Department	123 encounter_count
1 Emergency Department	553
2 General Pediatrics	194
3 PICU	138
NICU	115

4. Gold Layer

- Curated Delta Table
- Table: pediatric_encounters_gold
- Combined source fields with derived fields
- Ready for measures, visuals, and reporting

- Purpose: clean, validated table for semantic model**

pediatric_encounters_gold
ABC Patient_ID
ABC Encounter_ID
123 Age
ABC Age_Group
ABC Admit_Date
ABC Admit_Month
ABC Admission_Type
ABC Department
123 Length_of_Stay_Days
ABC LOS_Category
ABC Long_Stay_Flag
ABC Discharge_Status
123 Readmit_30_Day_Flag
ABC Readmit_Status

5. Power BI Output

- Built one-page Power BI report
- Created KPI cards and visual summaries
- Used DAX measures from semantic model

- Purpose: communicate validated encounter metrics in executive-ready report**



Department Summary

Department	Encounters	Avg LOS
Emergency Department	553	0.28
General Pediatrics	194	2.85
PICU	138	11.53
NICU	115	17.17

Repository Contents

The repository includes source and curated CSV files, screenshots, Power BI report export, Fabric workflow export, DAX measure documentation, data dictionary, validation summary, and reproducible Python scripts for generating and validating the Gold table.